



Get Started with the CDM for Digital Regulatory Reporting (DRR)

Module G – DRR and Use Cases

November 2024

Click below to get started

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This Get Started guide covers CDM for Digital Regulatory Reporting:

A	Introduction to the CDM	15 minutes	Complete
B	Accessing the CDM	10 minutes	Complete
C	How the CDM works	30 minutes	Complete
D	How the CDM is used	45 minutes	Complete
G	Introduction to DRR and Use Cases	30 minutes	Start

- In the past four modules, we have provided you with foundational knowledge on the CDM, how it works and how it is used. However, most of this has been generic, that is, not specific to any business products or domains.
- We will now turn to the regulatory reporting domain to introduce Digital Regulatory Reporting (DRR) followed by specific use cases to demonstrate how the CDM can be leveraged for regulatory reporting across jurisdictions.

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DRR topics and use cases covered in this module.

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Reduces regulatory reporting implementation and maintenance costs and mitigates non-compliance risk

Situation

Regulators around the globe are updating their reporting requirements to incorporate globally agreed data standards to improve the cross-border consistency of what is reported. But continued variations in the requirements, compounded by inconsistent industry interpretation and implementation, means errors and inconsistencies will continue, resulting in possible regulatory penalties for those firms that get it wrong.

Complication

Multiple regulators plan to implement new or revised rules, but they won't be identical. This means firms can't take the work completed for one jurisdiction and apply it to another. Instead, they need to individually interpret, apply and maintain each rule set, adding to costs, complexity and the burden on resources. Firms may also not interpret the requirements in the same way, leading to errors and the potential for regulatory censure.

Solution

ISDA's Digital Regulatory Reporting (DRR) solution:

- Creates consistency, as it is based on an industry-agreed interpretation of the rules and best practices and validates reporting by firms.
- Uses the Common Domain Model (CDM) to transform that consensus interpretation into unambiguous, machine-executable, open-access code.
- Can be extended to cover new rules and adapted to reflect future rule changes.
- Can be used for submission to trade repositories in multiple formats, including ISO20022.

Benefits

Avoids regulatory non-compliance: Regulators globally have issued fines of more than \$150 million for incomplete or inaccurate reporting. The ISDA DRR mutualized industry interpretation reduces this risk by ensuring all firms report the same data requirements in the same way.

Reduces costs: Using free machine-executable code produced from an industry mutualized approach and updated over time avoids duplication of effort across multiple industry participants. This enables fewer resource requirements per firm and allows resources to be reduced or reallocated to more value-add tasks.

- Digital Regulatory Reporting is a ground-breaking, industry led RegTech collaboration program.
- A number of ISDA member firms are “**digitizing**” (i.e. coding) the reporting logic and sharing that code with others in open source. The result is a standardized set of digitized representation of trade reporting rules and industry practices facilitated by ISDA.
- **Multiple firms are participating**, including global banks, asset managers and trade repositories.
- **DRR uses the CDM** to represent the transaction inputs into the reporting process as CDM “Events”.



REGULATORS & TRADE ASSOCIATIONS

Publish rules and develop best practices



Regulatory Text Artefacts



Tables of Data Field Mappings



Graphs and Spreadsheets of Trading Scenarios



Dealer X



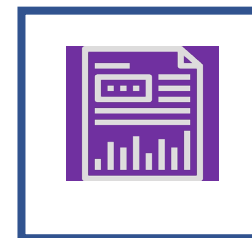
Dealer Y



Vendor Z

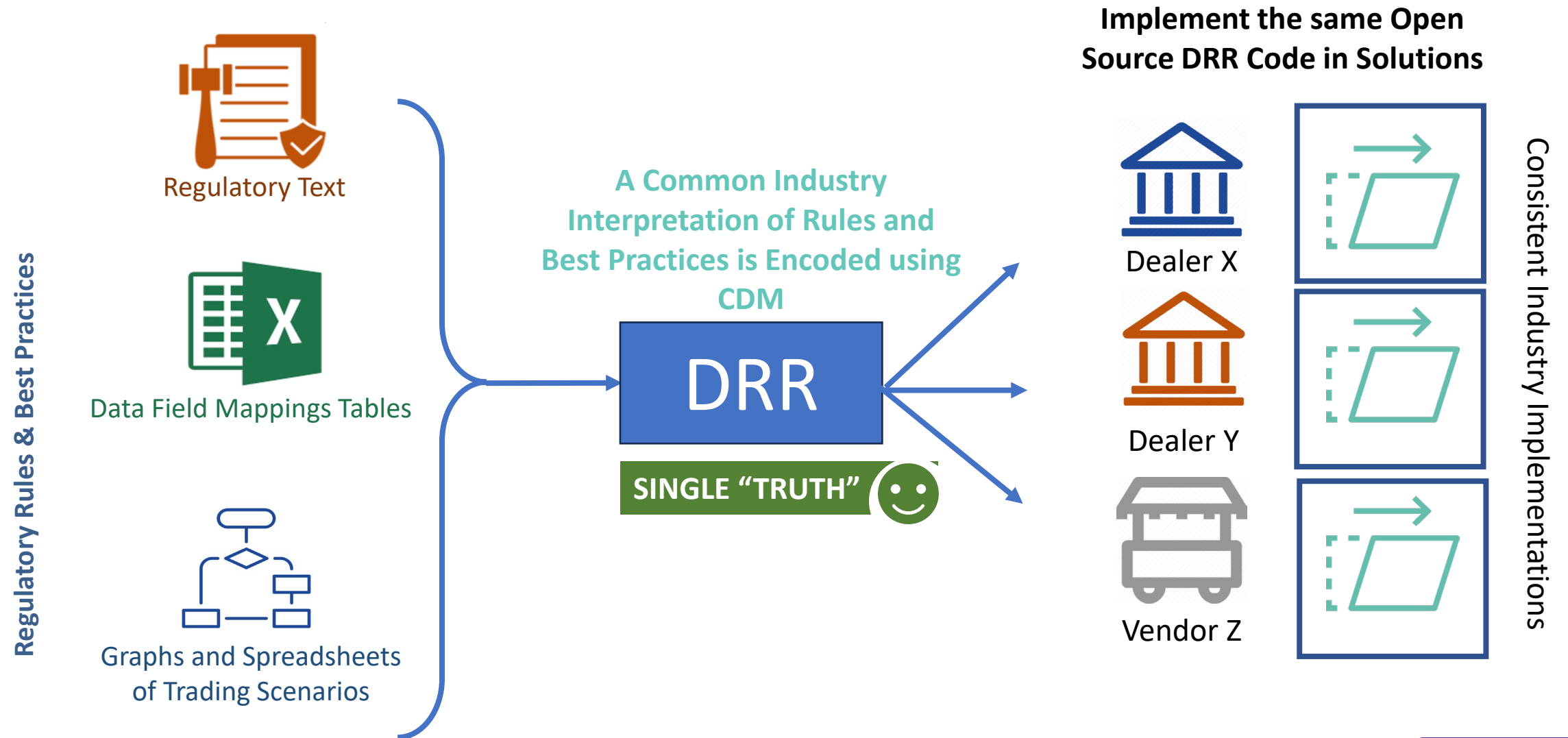
INDUSTRY PARTICIPANTS

Implement their own solutions based on individual interpretations



Resulting in fragmented and inconsistent implementations & operational inefficiencies

- Every industry participant left to implement their own version based on interpretation of artefacts
- Loss of interoperability between solutions
- Pervasive reconciliation issues and other operational inefficiencies



Mutualize regulatory reporting compliance effort

- Rule interpretations and compliance effort is spread across the industry

Gives you an unambiguous rule interpretation

- Reflects rules, guidance and industry best practices in an unambiguous way within the DRR model

DRR is open-access and increases transparency

- The DRR will be accessible to regulators and market participants

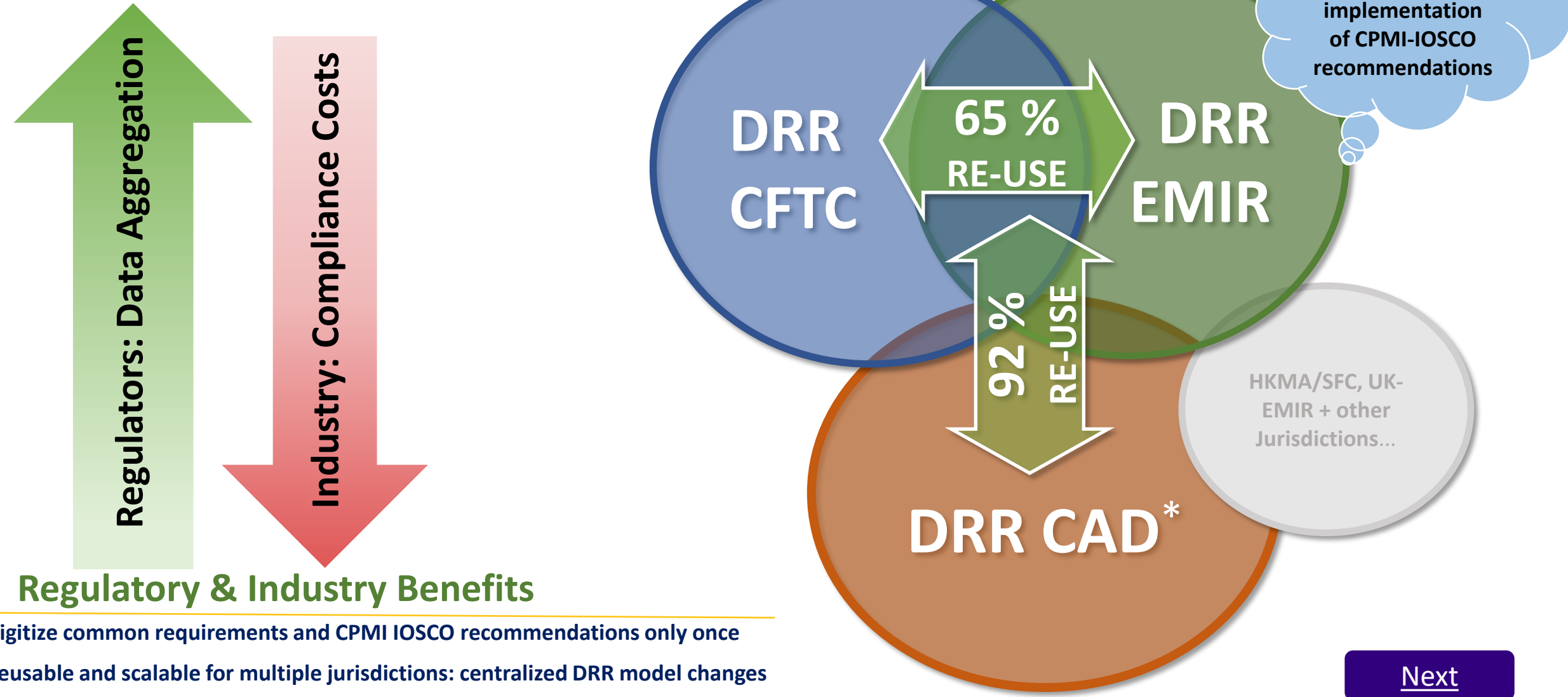
Defines core regulatory reporting ruleset only once

- Thereafter, only incremental efforts are required to extend the DRR model to other jurisdictions and future changes to reporting rules
- And such updates will be delivered through centralized DRR model changes

Significant resource and cost savings

- Through the mutualized effort, firms leveraging DRR using the CDM will reap significant compliance, reporting and implementation project savings

- DRR is re-usable and extensible to facilitate consistent implementation



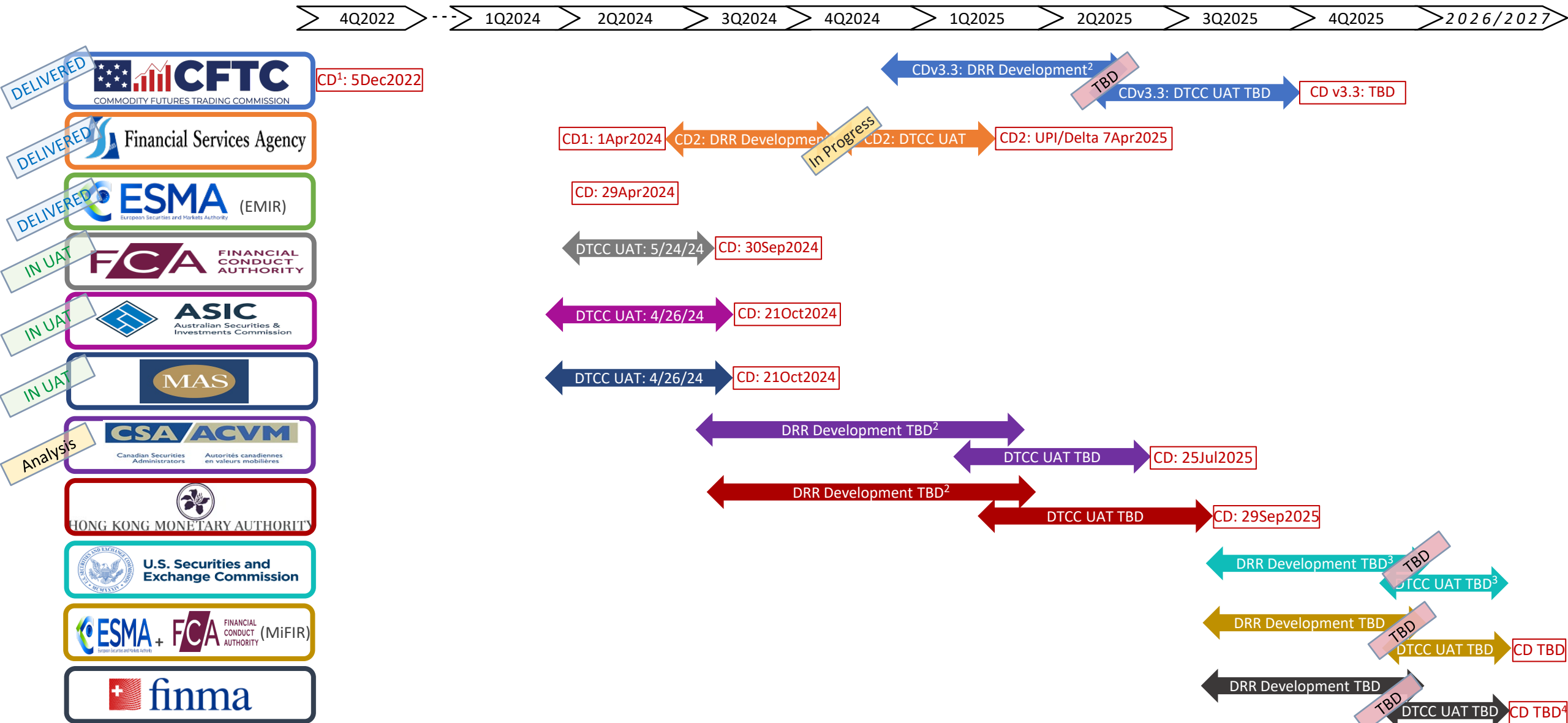
Digitize common requirements and CPMI IOSCO recommendations only once

Reusable and scalable for multiple jurisdictions: centralized DRR model changes

* Canada jurisdiction initial assessment of overlap of field definitions, format, allowable values with CFTC+EMIR: 77% full; 15% partial.

[Next](#)

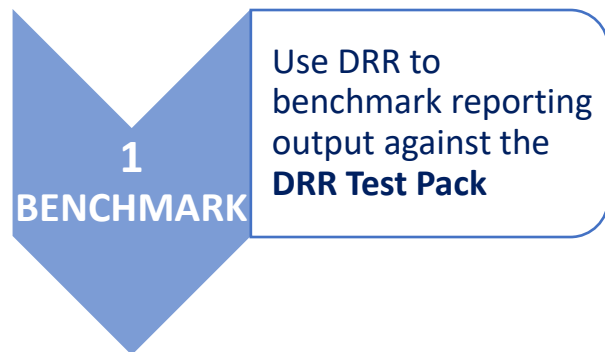
G1.5 Background to DRR: Delivery Status and Progress (as of November 2024)



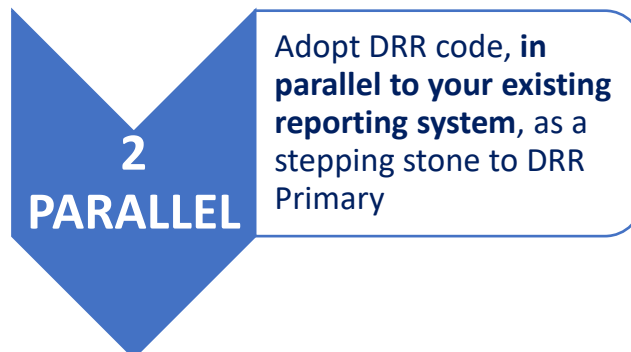
1) CD=Compliance Date. 2) Timing dependency on timing of potential DRR refactoring. 3) DRR timing & need for UAT dependent on SEC action re: Nov2025 relief expiry. 4) expected 2027 (subject to change).
Please note, this timeline is as of November 2024 and will be updated as required.

G1.6 Digital Regulatory Reporting – Different Forms of Adoption

DRR AS BENCHMARK



BUILD DRR INTO YOUR SYSTEM



VALUE

- **DRR is a mutualized community solution.** It is the industry's **common interpretation of reporting requirements** and offers a **maintained reporting benchmark** for firms
- Human readable '**golden source**' ensuring **alignment with industry consensus**
- Offers confidence in **reporting accuracy**
- Serves as a **control or corroboration** of firms' reporting output to trade repositories
- Can be used by vendors for **accuracy check** services
- **Traceability** of reported data back to the 'golden source'
- **Can be used by vendors as a 'utility'** supporting their services particularly **reporting software** and **accuracy checks**
- **Delegated reporting services** aligned with industry standards

IMPLEMENTATION

- Light, may require development of translation from firms' data formats to those supported by publicly available CDM & DRR
- Free access to Rosetta community edition needed for DRR and CDM and/or connection via API
- Implemented either directly through internal integration effort or through use of a third-party vendor who has adopted the DRR code. Requires adoption of the Common Domain Model (CDM) for data staging
- Development and deployment of a run-time execution engine, integration of DRR code artefacts, translation from internal data formats to CDM, own code generator based on publicly available formats (distributed with CDM/DRR)
- Partial adoption of CDM as Proof-of-Concept
- Adoption of the CDM for data staging (natively or via vendor)

Q: Let's check your understanding of Digital Regulatory Reporting. Which of the following statements is most accurate about the advantages it provides?

Click on the answer
you think is correct

A1 Significantly lowers effort and cost to implement changes to rules.

Incorrect

A2 Reduces pairing and matching issues.

Incorrect

A3 Facilitates consistent implementation of global trade reporting requirements.

Incorrect

A4 Anticipated improvement in quality of reported data for regulatory use and analysis.

Incorrect

A5 All of the above.

Correct

G

DRR topics and use cases covered in this module.

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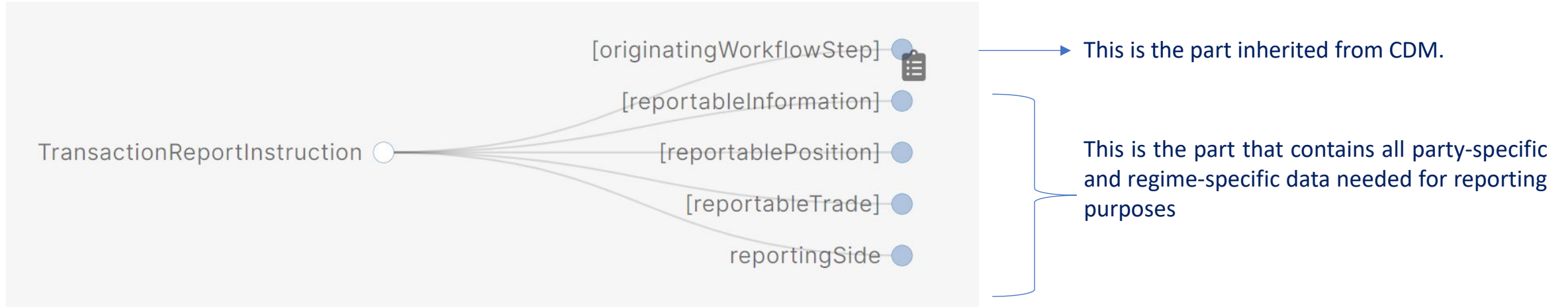
5

Get Involved

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The Digital Regulatory Reporting is a model that extends CDM, adding regulatory reporting features to it. Therefore, DRR leverages CDM in order to generate and validate reports from business events presented in a CDM format. Please ensure you have completed modules A-D covering CDM basics before progressing further.

In terms of data model, DRR adds a higher layer of attributes that contain all the needed information to generate a well-formed and valid report.



Currently, DRR extends the **ReportableEvent**, creating the **TransactionReportInstruction**. This type is the root element of the DRR and it is from where all reports are generated. This means that the JSON input file of a report generation needs to be a **TransactionReportInstruction**. This type contains a feature needed to specify delegated reporting.

DRR conforms to the CDM's design principles where data and functions are used to model reportable fields and logic.

Different Models can be built on CDM with alternative governance or licensing frameworks as appropriate

Core CDM under open source license and governance at FINOS

CDM

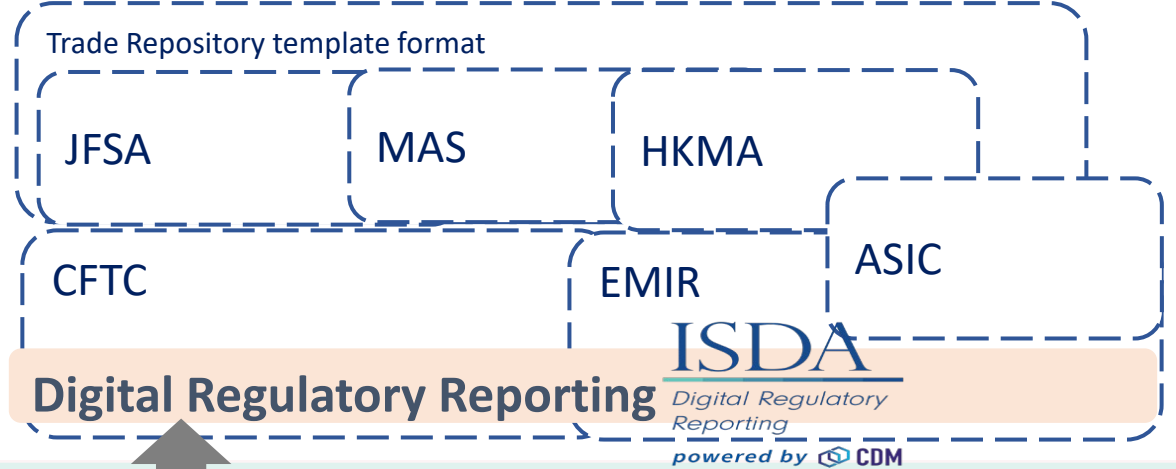
product – the product model

observable - asset & event (e.g. credit event) – basic building blocks to construct products

base - common elements: date & time, static data etc.



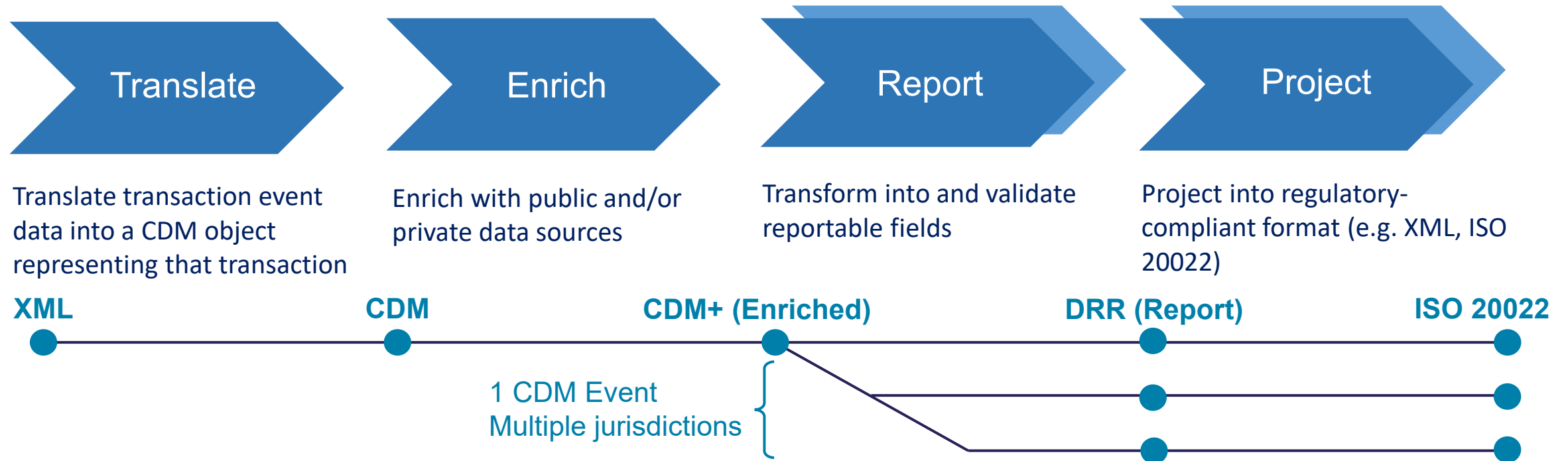
Maintainers:



[Next](#)

From end to end in four key steps

At each step, business logic is expressed and can be leveraged as part of an application layer. As part of the DRR project, ISDA is collaborating with members to review, test and project to ISO 20022 (or any other relevant reporting format for jurisdictions). Synthetic data samples for each step's input and output are captured in a "Test Pack" allowing validation for any implementation against those tests.



Anatomy of a reporting rule in DRR:

A reporting rule binds two components together:

- The functional (i.e. “machine-executable”) **logic**
- A specific reference to a **document** supporting that logic (e.g. regulatory text, guidance etc.)

DRR uses the MER Rosetta DSL to enable regulatory experts to read / write functional logic reflecting their interpretation of regulatory rules.

FixedRateLeg1
CFTC Part45 x

Type
Report Field

Name
FixedRateLeg1

Reference
 appendix 1
 dataElement 67
 field Fixed Rate 1
 footnote 54

Provision
 The allowable values are restricted based on CFTCs jurisdictional requirements.

Rule Definitions

FixedRateLeg1
 extract
 TradeForEvent(...)

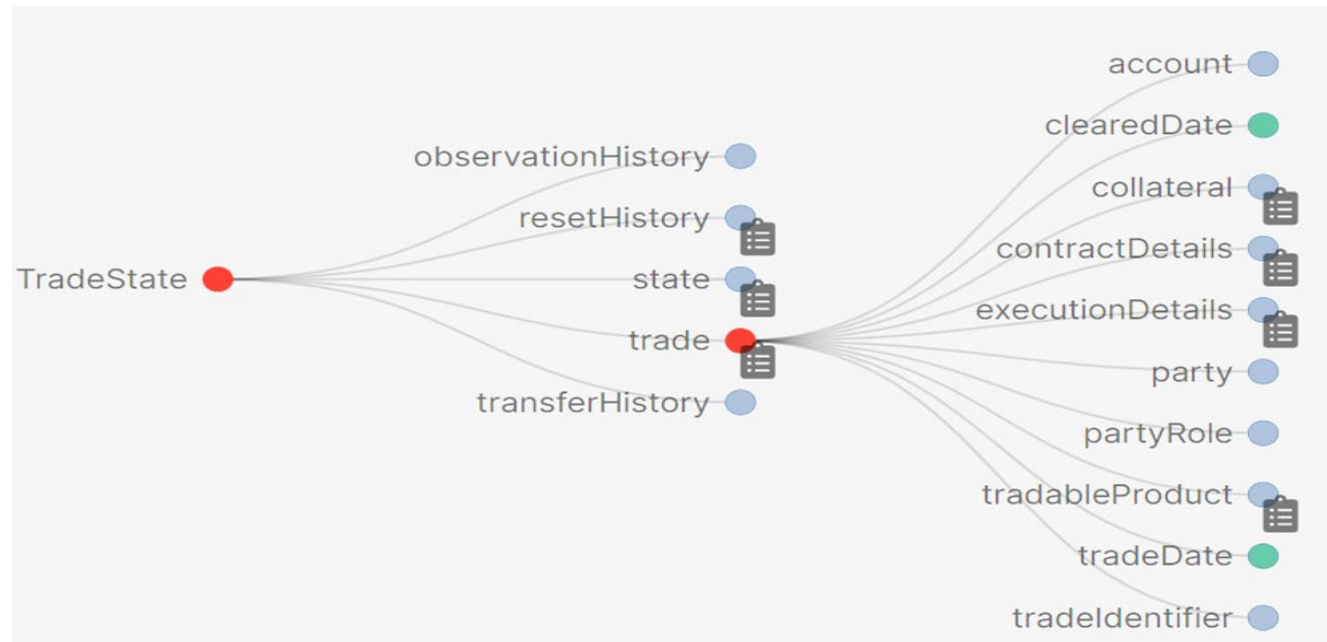


```
reporting rule FixedRateLeg1
[regulatoryReference CFTC Part45 appendix "1" dataElement "67" field "Fixed Rate 1"
  provision "For each leg of the transaction, where applicable: for OTC derivative
    transactions with periodic payments, per annum rate of the fixed leg(s)."]
[regulatoryReference CFTC Part45 appendix "1" dataElement "67" field "Fixed Rate 1" footnote "54"
  provision "The allowable values are restricted based on CFTCs jurisdictional requirements."]
extract TradeForEvent( ReportableEvent ) then extract ProductForTrade( Trade ) then extract
  CDEInterestRateFixedRate( InterestRateLeg1( Product ) ) as "67 Fixed rate-Leg 1"
```

As you may remember from Module C ('How the CDM Works'), a **BusinessEvent** in CDM is used to apply an **Instruction** on a **TradeState**.

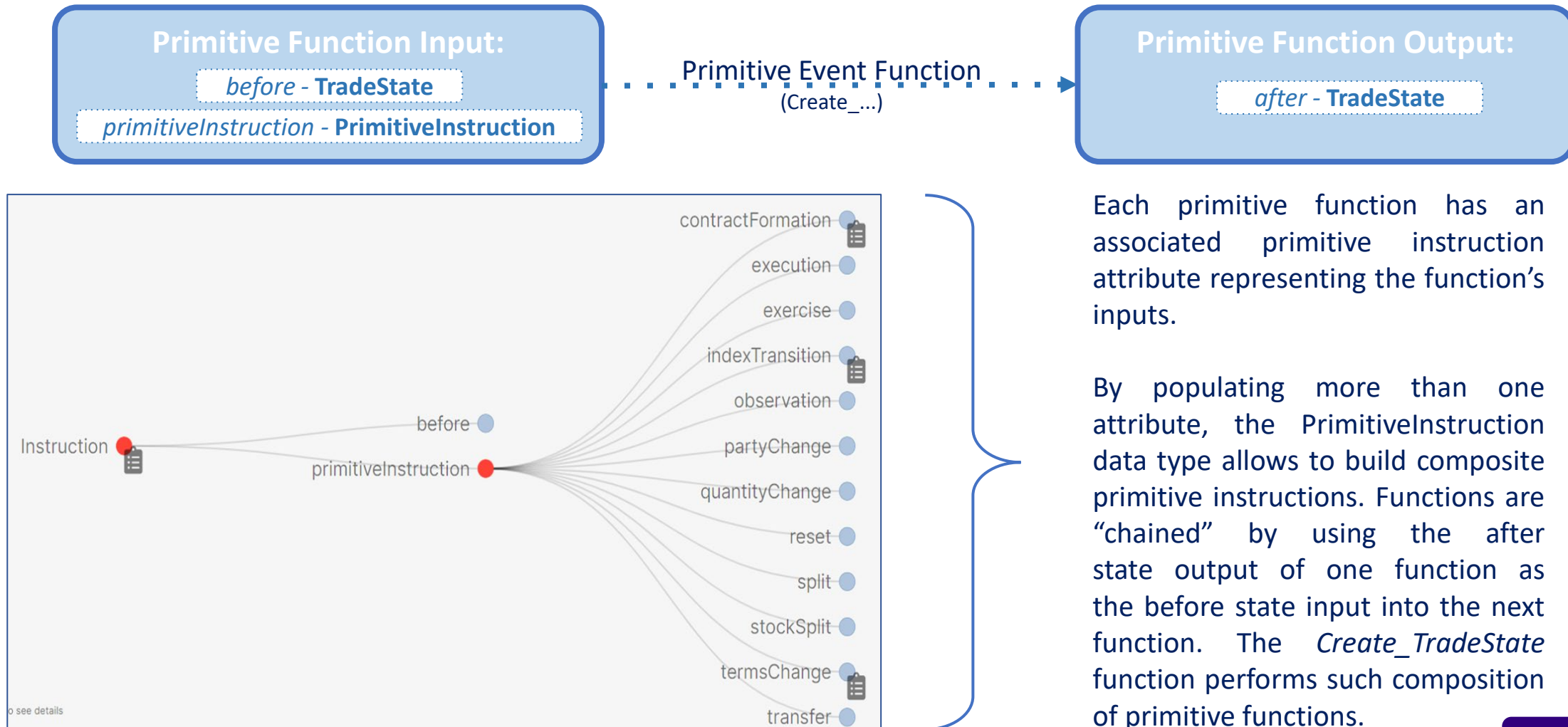
TradeState:

- Defines the fundamental financial information that can be changed by any life-cycle event.
- It is the input and output of any state transition - with each event, a new TradeState instance is created.

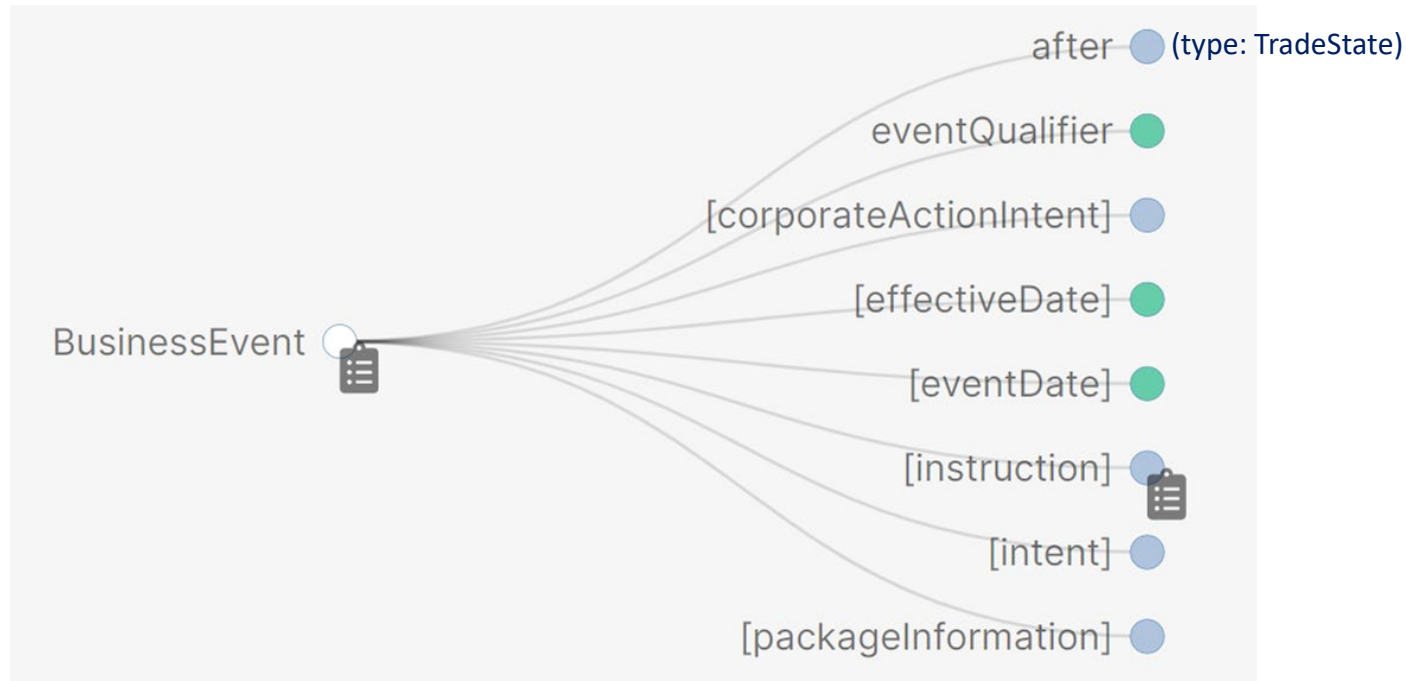


G2.6 How DRR is Built on CDM - Components: Instructions

- Contains the information that it is used by the *Primitive Functions* to perform the next TradeState(s) after an event has occurred.



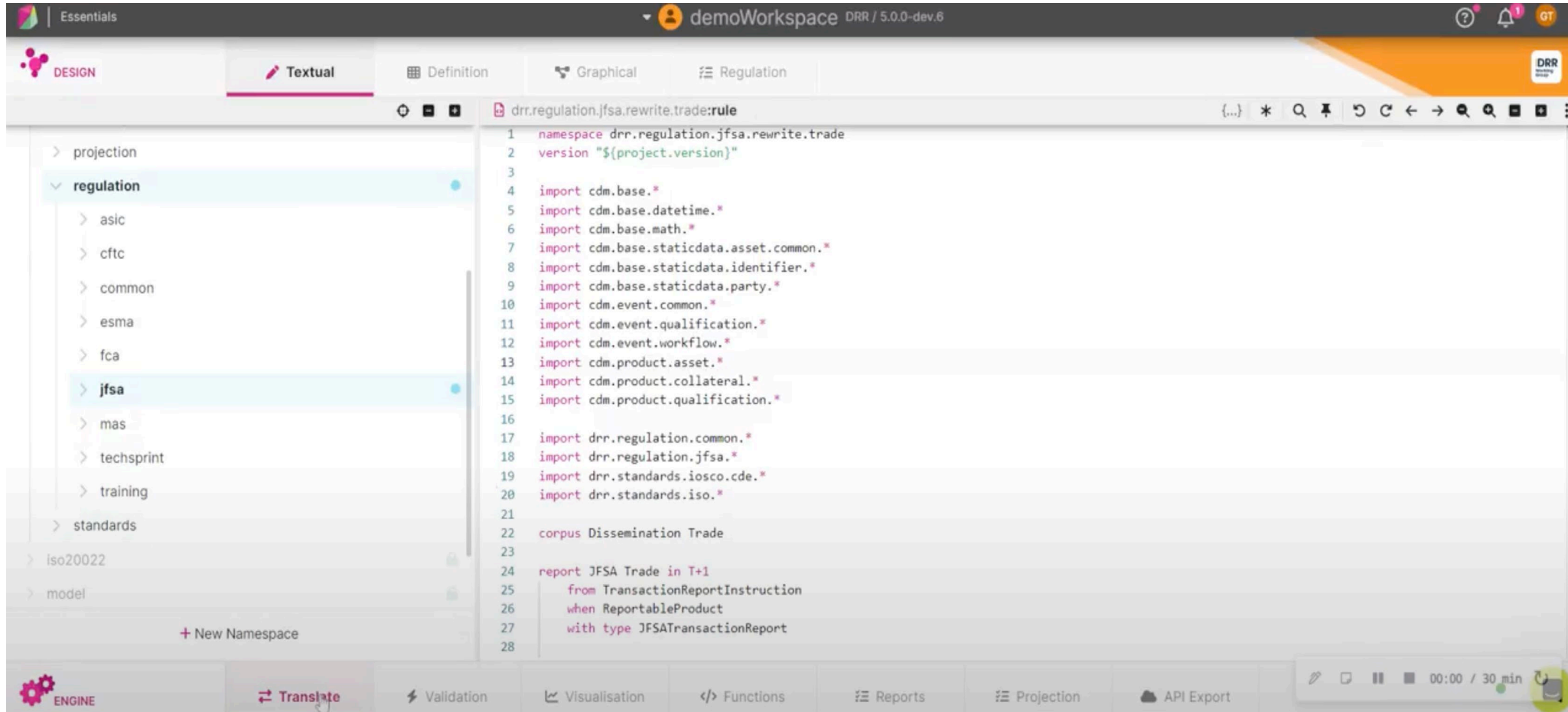
The Business Event represents the **current life cycle event** of a trade. It contains the resulting TradeState after an event has occurred, as well as the instructions that generate it.



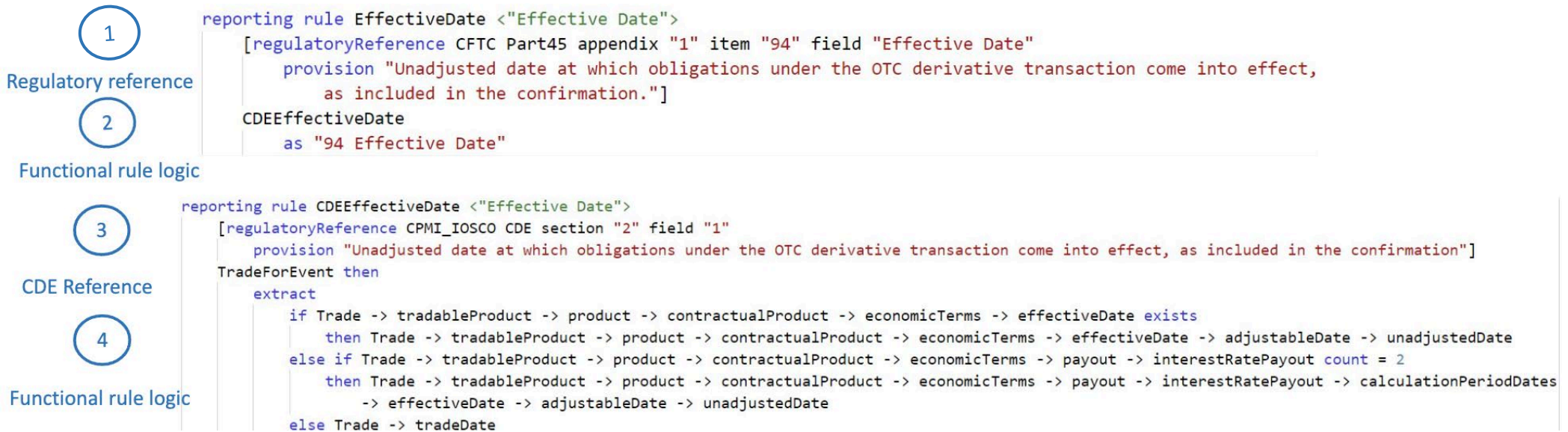
The Business Event is specified functionally by the *Create_BusinessEvent* function.

A business event is atomic, i.e. its primitive components cannot happen independently. They either all happen together or not at all.

[Business Events – Demo Video](#)



- Reporting rules are defined as logical instructions (i.e., functions) applied to CDM data objects in **human readable, machine executable, open-source code**.
- Each rule is associated to *regulatory references*: rules, other guidance, or best practices, etc.



For further examples and details see: https://www.isda.org/a/KFXTE/DRR-Paper-Updated-Audio-2020.mp4?_3

- The below sample outlines an EU-EMIR reporting rule's representation in DRR for the field Report tracking number. The DRR model applies filtering to obtain the required data element contained in the reportableInformation.

```
reporting rule ReportTrackingNumber from TransactionReportInstruction: <"Report Tracking Number">
[regulatoryReference ESMA EMIR RTS table "2" dataElement "2" field "Report Tracking Number"
  provision "Where a derivative was executed on a trading venue, a number generated by the trading venue and
    unique to that execution."]
filter IsAllowableAction
then extract if
  IsEU(GetOrFetchMicData(
    reportableInformation -> enrichment -> micData,
    GetVenueOfExecution(reportableInformation)
  ) -> countryCode
  ) = True then GetReportTrackingNumber
as "2.2 Report Tracking Number"
```


- The DRR model specifies the application of certain rules. The example below applies the industry agreed approach of not defaulting blank values for JFSA reporting.

```

1609 reporting rule TotalNotionalQuantityLeg2 from ReportableEvent: <"Total notional quantity-Leg 2">
1610     [regulatoryReference JFSA Trade dataElement "97" field "Total notional quantity"
1611         provision "For each Leg of the transaction, where applicable: aggregate Notional quantity of the underlying asset of leg 2."
1612         Where the Total notional quantity is not known when a new transaction is reported, the Total notional quantity is calculated as follows:
1613+    [regulatoryReference JFSA Trade dataElement "97" field "Total notional quantity" footnote "1"
1614+        rationale "Peer Review Group members agreed DRR rules should not default the field value."
1615+        rationale_author "Peer Review Group - 05/10/2022"
1616+        provision "Note: '999999999999999999999999' is accepted when the value is not available at the time of reporting."]
1617 // todo: add coverage for equity swaps
1618 extract TradeForEvent
1619 then extract ProductForTrade
1620 then extract
1621     if IsCommodity
1622     then CDECommodityTotalNotionalQuantity(CommodityLeg2)
1623     as "97 Total notional quantity-Leg 2"

```

Q: Which of the following statements is true about rules in DRR?

Click on the answer
you think is correct

A1 Firms can code their own rule interpretations into DRR.

Incorrect

A2 Rules do not need to be associated with regulatory references.

Incorrect

A3 Regulatory references are the only source used for rule interpretation.

Incorrect

A4 Each rule is associated to rules, other guidance, or best practices.

Correct

A5 None of the above.

Incorrect

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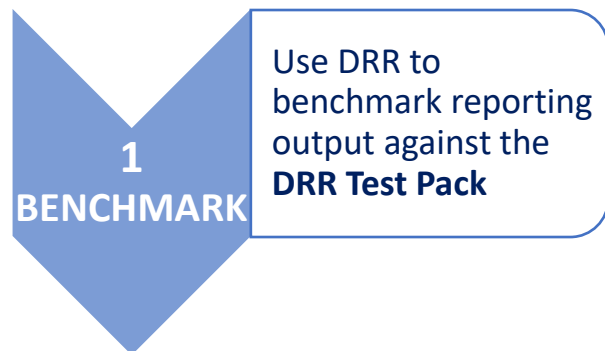
[Start](#)

5

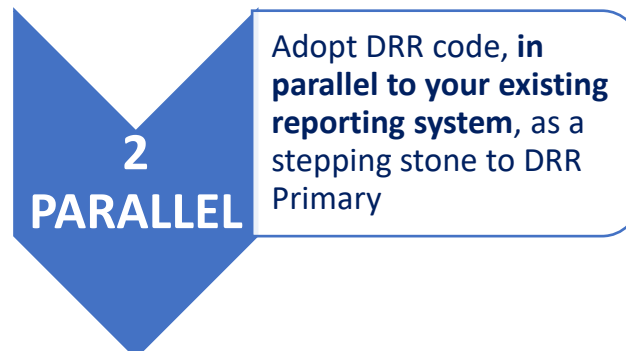
Get Involved

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DRR AS BENCHMARK



BUILD DRR INTO YOUR SYSTEM



VALUE

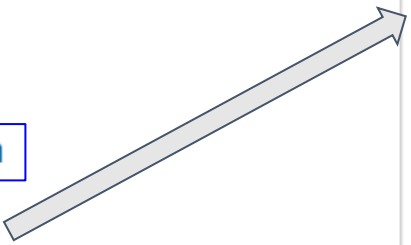
- **DRR is a mutualized community solution.** It is the industry's **common interpretation of reporting requirements** and offers a **maintained reporting benchmark** for firms
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- Partial adoption of CDM as Proof-of-Concept
- Adoption of the CDM for data staging (natively or via vendor)

DRR can be implemented into a firm's reporting system for the purpose of generating outgoing reports to TRs.

```
report JFSA Trade in T+1
from TransactionReportInstruction
when ReportableProduct
with type JFSATransactionReport
```



In DRR, reports are defined in four steps: **report name**, **input**, **eligibility** and **report type**.

When inputting a reportable transaction and selecting a report, the reporting type gets called. This reporting type is built by a set of attributes (such as the CDM ones, with its name, its type and its cardinality) that contain a reference to a **reporting rule**.

The Reporting rules are coded to get an input message (a TransactionReportInstruction) and extract from it a particular information. For instance, the reporting rule **EffectiveDate** will extract the effective date of the trade. The data extracted will be used to populate each attribute of the reporting type, consequently generating a full report once all reporting rules have been called.

Despite the capabilities of the DSL to apply eligibility criteria in the report generation, the implementation of the eligibility rules is beyond the model's scope and has not been included yet.

```
type JFSATransactionReport:
[rootType]
effectiveDate ISODate (0..1)
| [ruleReference EffectiveDate]
expirationDate ISODate (0..1)
| [ruleReference ExpirationDate]
earlyTerminationDate date (0..1)
| [ruleReference EarlyTerminationDate]
reportingTimestamp zonedDateTime (1..1)
| [ruleReference ReportingTimestamp]
executionTimestamp zonedDateTime (0..1)
| [ruleReference ExecutionTimestamp]
entityResponsibleForReporting LEIIdentifier (1..1)
| [ruleReference EntityResponsibleForReporting]
counterparty1 LEIIdentifier (1..1)
| [ruleReference Counterparty1]
counterparty2 Max72AlphaNumericText (1..1)
| [ruleReference Counterparty2]
counterparty2IdentifierType boolean (1..1)
| [ruleReference Counterparty2IdentifierType]
direction1 Max4Text (0..1)
| [ruleReference Direction1]
```

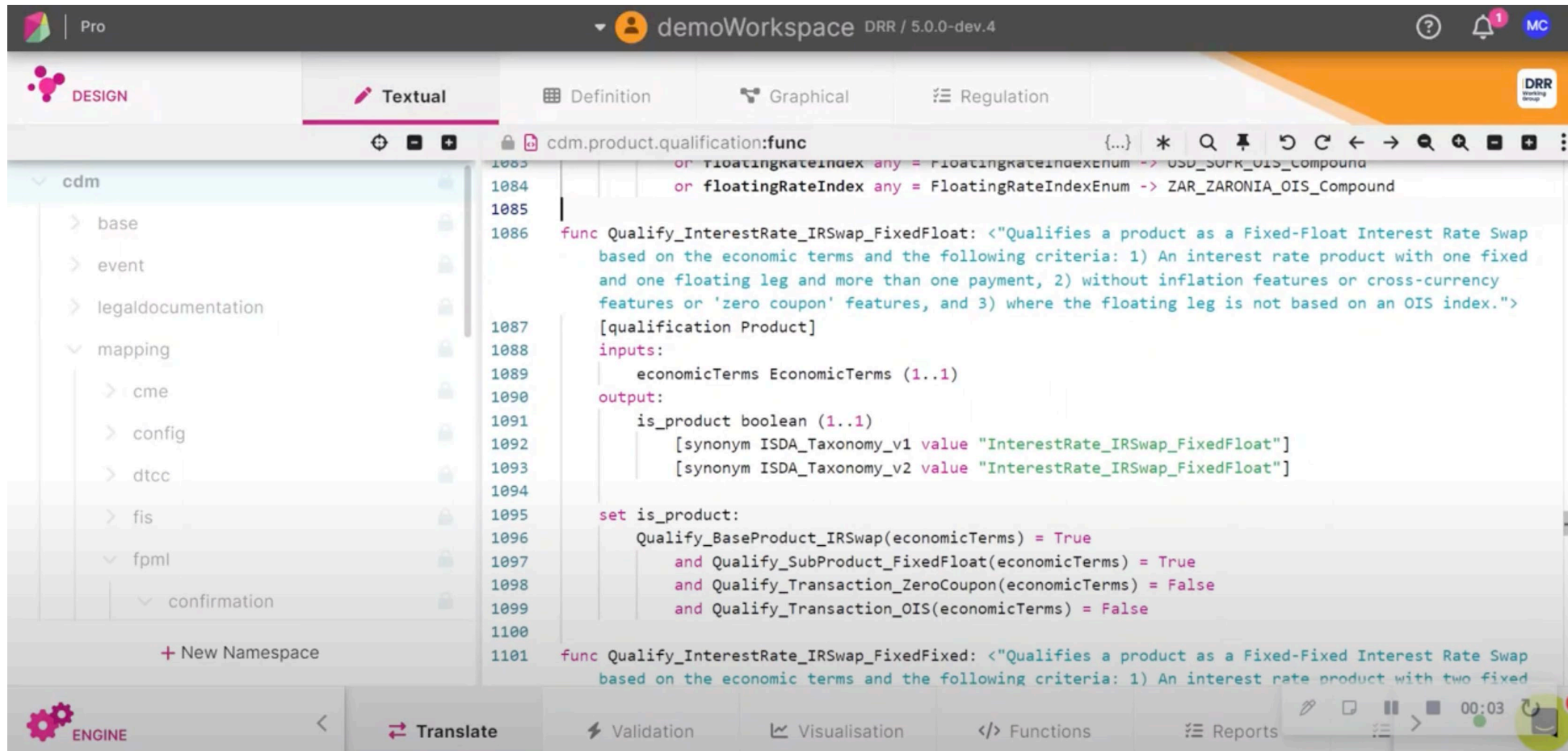
G3.2 Reporting Rules Syntax

The reporting rules are defined in the namespaces with the suffix “:rule” and they are introduced by the keyword “reportingRule” together with the **validation rule name and the starting point of the extraction**. This is followed by the **regulatoryReference**, the **functional expression** that extracts the value to be reported, and finally the **“as” keyword setting the label onto the value to appear as the column name in the report**. They may contain **plain-text descriptions**.

```
reporting rule EventTimestamp from TransactionReportInstruction: <"Event Timestamp">  
[regulatoryReference JFSA Trade dataElement "104" field "Event Timestamp"  
  provision "Date and time of occurrence of the event as determined by the reporting counterparty or a service  
    provider.  
    In the case of a clearing event, date and time when the original swap is accepted by the derivative clearing  
      organization (DCO) for clearing and recorded by the DCOs system should be reported in this data element.  
    The time element is as specific as technologically practicable."]  
  extract originatingWorkflowStep -> timestamp  
  then filter qualification = EventTimestampQualificationEnum -> eventCreationDateTime  
  then distinct only-element  
  then extract dateTime  
  as "104 Event timestamp"]
```

Functional expressions are composable, so a rule can also call another rule or function.

[Report Generation - Demo Video](#)



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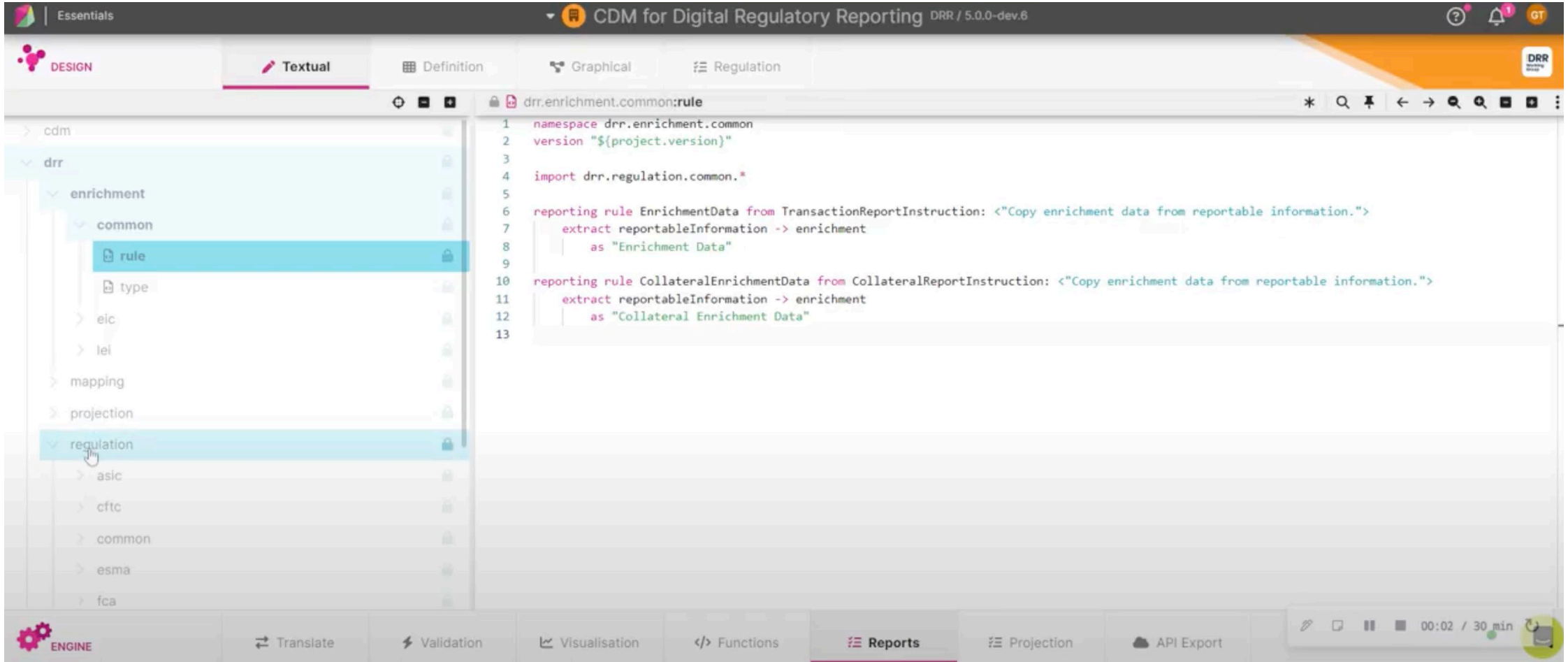
The report can be validated through type conditions, which are logical expressions associated to a data type, in this case the report. In alignment with the regulatory guidelines, they apply logical statements that evaluate to True or False.

As part of the report building process, all the conditions are evaluated and if any evaluates to false, the validation rule fails and a failure, with the corresponding error code, is displayed in the Reports Tab.

They are defined right after the report type and introduced by the “condition” keyword together with the **condition name or error code to be displayed**, the **docReference**, the **boolean expression**, and may contain **plain-text descriptions**.

```
condition DTCC_JFSA_BR_0024_02: <"Clearing receipt timestamp Condition 2">  
  [docReference JFSA DTCC_Specs dataElement "24" field "Clearing receipt timestamp"  
    | provision "Required if: Cleared = Y AND [Action Type] = 'TERM' AND [Event Type] = 'CLRG' or = 'CLAL'"]  
    if cleared = "Y" and actionType = "TERM" and ["CLRG", "CLAL"] any = eventType  
      then clearingReceiptTimestamp exists
```

[Report Validation - Demo Video](#)



The screenshot displays the CDM for Digital Regulatory Reporting (DRR) software interface. The top bar shows the title "CDM for Digital Regulatory Reporting DRR / 5.0.0-dev.6". The left sidebar contains a tree view with the following structure:

- cdm
 - drr
 - enrichment
 - common
 - rule** (selected)
 - type
 - eic
 - lei
 - mapping
 - projection
 - regulation** (selected)
 - asic
 - cftc
 - common
 - esma
 - fca

The main editor area shows the definition for the selected "rule" in the "regulation" section. The code is as follows:

```
1 namespace drr.enrichment.common
2 version "${project.version}"
3
4 import drr.regulation.common.*
5
6 reporting rule EnrichmentData from TransactionReportInstruction: <"Copy enrichment data from reportable information.">
7   extract reportableInformation -> enrichment
8   as "Enrichment Data"
9
10 reporting rule CollateralEnrichmentData from CollateralReportInstruction: <"Copy enrichment data from reportable information.">
11   extract reportableInformation -> enrichment
12   as "Collateral Enrichment Data"
13
```

The bottom bar contains several tabs: ENGINE, Translate, Validation, Visualisation, Functions, **Reports** (selected), Projection, and API Export. A status bar at the bottom right indicates the time "00:02 / 30 min".

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- We have already seen that the Common Domain Model (CDM) is an open source project, overseen by FINOS. DRR is open to consume however it is distributed under an ISDA license.
- DRR specific working groups are therefore run by ISDA.

Here are the key ways to interact with the CDM community:

- Working Groups
 - These are the primary interactive forums for participants to collaborate on the development of the CDM. With defined objectives and topic areas, the Working Groups meet on a regular, defined schedule.
 - Email help@finos.org to be added to meeting invites directly, or find a meeting in the [FINOS Community Calendar](#).
- Email
 - FINOS maintains a number of mailing lists to distribute information on the CDM and also to enable collaboration within the community, including the Working Groups.
- GitHub
 - As we have seen, [GitHub](#) is the repository that stores the CDM model.
 - But it is also used for communication, including tracking issues and model updates, and the meeting schedules, agendas and minutes for the Working Groups.

The below ISDA Working Groups are specific to Digital Regulatory Reporting and are open to [ISDA members](#) only.

DRR Steering Committee:

Determines the roadmap including the scope, priorities and timeline for DRR development. The Steering Committee meets bi-weekly and is comprised of institutions with reporting obligations under a particular set of trade reporting rules who intend to adopt the DRR.

EU/UK/North America Peer Review and APAC/JFSA Peer Review Groups

These groups review the code completed by digitisers and validate that coding of a rule is consistent with reporting expectations. The Peer Review groups are open to all and meet weekly or bi-weekly – there are no minimum requirements to join this group, although we encourage group members to actively engage in the review of the DRR code. Hover over and click the names in the title to join on ISDA's Committees page.

How to Join ISDA DRR Working Groups:

Account Registration

1. click the [Sign Up Now](#) link located in the top right corner of [our homepage](#).
2. Look for an email from isdams@isda.org, which contains instructions for activating your account.
3. If you do not receive the activation email within 1 hour, please contact isdams@isda.org for assistance.

Working Groups Registration

1. Access your [Committee Dashboard](#).
2. Add yourself to the ISDA working groups and distribution lists you wish to join.

A	Introduction to the CDM	15 minutes	Complete
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D	How the CDM is used	45 minutes	Complete
G	Introduction to DRR and Use Cases	30 minutes	Complete

Well Done! You have completed Module G.

You can review any of the complete sections again or move on to the next module covering Advanced Features.

The below items point to key documents and tools leveraged as part of the CDM project development and implementation.

Introduction to DRR

- [Video: Introduction to ISDA's Digital Regulatory Reporting Initiative – International Swaps and Derivatives Association](#)
- [ISDA White paper on Digital regulatory reporting](#)
- [DRR Fact Sheet](#)
- [DerivatiViews article](#)

Key links on project modelling tools

- [Rosetta platform](#)
- [DRR documentation](#): Documentation on all the key components of the DRR model
- [CDM Documentation](#): Documentation on all the key components of the CDM model.
- [CDM repository](#): Source code repository for the CDM, which is now hosted at FINOS